

SYLLABUS FOR CERTIFICATE COURSE IN COMPUTER HARDWARE

Unit	Objective	Learning Outcomes	Subject Content	Teaching & Learning Methods	Time (house)	
					Theory	Practical
1	✓ <i>Have an overall idea of Evolution of Computers and Basic Fundaments and Concepts with regard to Computer Hardware</i>	- Introduces the Computer History - Explains fundamentals of Modern Computers	PC & Laptop Fundamentals - Evolution of Computers from Vacuum tube to present - Basic Computer Concepts	- Lecture discussions - Video Demonstrations - Group Presentations	03	03
2	✓ <i>Build up a fully competent person in Installing, configuring and maintenance Operating System with the background knowledge and current market trends in Operating Systems</i>	- Analyses Operating System editions, versions & Features. - Chooses an Operating System according to the customer requirement and situation. - Installs, Configuring and Maintaining Computer Systems with full competency.	Operating Systems - Different types of Operating Systems - Develop communication skills and train to identify the best OS according to the requirements - Install, configure and maintain Operating Systems	- Lecture with Practical Demonstration - Video Demonstration - Hands on Practical	06	10
3	✓ <i>Provide competencies to install configure and optimize laptops</i>	- Describes difference types of Portable PCs - Disassembles Portable PCs and fix the problem.	Portable Components & Functions - Identify Laptops, Ultra book, Tablet -pc, Notebook, notepad - Disassemble portable PCs - Assemble portable PCs	- Lecture with Demonstration - Video Demonstrations - Physical Inspection of all parts - Group Presentations	05	10

4	✓ <i>Build an Experts in installing, configuring and Optimizing PCs</i>	- Explains PC types - Disassembles PCs	Identify PC & Assemble a PC - Identify Generations of Processors (i-3, i-5, i-7) - Identify parts of the system unit (Hard Disk, CD, DVD RW, Memory types, power supply)	- Lecture with Practical Demonstration - Video Demonstration - Hands on Practical	05	12
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5	✓ <i>Enhance abilities to Identify and configure the different types of Input /Output devices in Computer Hardware Perspective</i>	- Inquires Input/output devices - Configures the Input /Output Devices - Trouble shoots Input /Output Devices	Install and Configure Peripherals' - Different types of Input /Output devices - configure and trouble shooting	- Lecture with Demonstration - Video Demonstrations - Physical Inspection of all parts	03	08
6	✓ <i>Develop skills to Install drivers for all kind of PCs and Laptops</i>	- Discusses the Driver model numbers and manufactures - Installations Virus Guards and some application software's	Install Software and Driver – PC/Laptops' - Identify the PC and Laptop models - Identify the different kind of Chipsets - Identify the different kind of device drivers	- Lecture with Practical Demonstration - Video Demonstration - Hands on Practical	05	10
7	✓ <i>Empower Gain an overall knowledge on Electronics Components required for Computer Hardware Technician</i>	- Explains about Basic Electronic Components - Measures and Converting Electronic Values - Describes Electronic Measurement tools - Troubleshoots Devices	Basic Electronics - Electronic Components (Resistors, Capacitors, Diode, Transistors) - Basic Measurements - Troubleshooting Devices	- Lecture discussions with Guest Lecturer - Video Demonstration - Hands on Practical - Individual and Group activities	04	04
8	✓ <i>Train Troubleshoot all kinds of laptops and PCs in the market.</i>	- Discusses the issues in a system - Find solutions and choose the best. - Repairs and maintenance	Troubleshoot & Repair PC, Laptops - Error Detection - Solution options - Fixing the error and maintain	- Lecture with Practical Demonstration - Video Demonstration - Hands on Practical	03	10
9	✓ <i>Knowledge provide comprehensive Upgrade all kinds of software and</i>	Explains the Software and Hardware requirements	Upgrade software & Hardware Identify and track for new upgrades	Lecture with Practical	03	06

	<i>hardware with various technologies.</i>	Upgrades the hardware and software according to new technologies	Identify Upgrading methods and schedule and perform upgrades	Demonstration - Video Demonstration - Hands on Practical		
10	✓ <i>Enhance abilities to configuring a network system Networking concepts and to obtain a complete</i>	- Analyses network types (topologies) - Creates and Maintains Network System - configures the network	Introduction to Network and Setup, Maintain computer Network - Identify the network areas and topologies - Identify the network tools - configure and maintain the network	- Lecture with Practical Demonstration - Video Demonstration - Hands on Practical	05	10
11	✓ <i>Develop skills on modern Cloud Concepts</i>	- Introduction to cloud storage - Describes different ways of clouds	Cloud Concepts and Technologies Introduction to store data in cloud	- Lecture with Practical Demonstration - Video Demonstration - Hands on Practical with the assistance of the lecturer	03	03
12	✓ <i>Facilitate to experience the virtual environment and to work with the real-time</i>	- Previous to Hyper -V platform - Configures hyper – v - Troubleshoots Hyper -v	Virtualization Technology Establish a Network between the host – pc and Virtual PC	- Lecture with Practical Demonstration - Video Demonstration - Setting up Hyper-V environment	04	04
13	✓ <i>Train to deal with market orientation of computer hardware.</i>	- Analyses current market trends - Evaluate the customer requirement and provide best solution for the customer with the knowledge of current market trends	Market Orientation - Identifying the market by practically participating for the Orientation Programme at the current market - Analyze the customer requirements - Preparing quotations. - Attend to after sales services	- Lecture with Practical Demonstration - Video Demonstration - Hands on Practical with the assistance of the lecturer	04	07
					53	97
						150hrs

METHOD OF ASSESSMENT

Final Evolution Certificate Course in Computer Hardware - Knowledge and the Skills of the students are assessed in the following manner at the completion of the Course. However, continuous assessments are done at the completion of each Unit which can be written or practical appropriate to the Unit.

1. Written Papers

There are 2 papers comprise as follows;

- 1.1. **Paper I** - Multiple Choice Question (MCQ) Paper comprises of 40 Nos. MCQ questions. (40 Marks) – One hour
- 1.2. **Paper II** – 8 Essay type Descriptive Questions out of which students must answer 6 questions (60 Marks) – Two hour

2. Practical Test

Practical based on actual situation. Students are given an actual customer requirement and will be questioned to provide total solution from providing quotation to delivering fully set up computer system to the customer. This question will be based on the following;

- Understanding customer requirements
- Providing Quotations and convincing customer on same
- Assembling and Software Installation
- Troubleshoot any errors arises
- Questions based on after sales services

Method of Assessment	Weighting
Written Papers	40%
Practical Test	60%

Final result / Grades will to decide by the national youth services council on the final marks obtained for written papers and practical test.